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Opportunities and challenges of machine learning in bioprocesses: Categorization from different perspectives and future direction

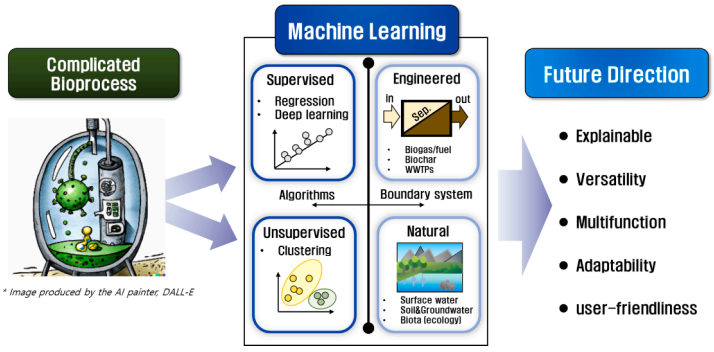
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HIGHLIGHTS

- We outline machine learning applications to bioprocess from different perspectives.
- Categorization was made per the purpose of the ML (supervised vs unsupervised).
- Categorization was made per the system boundary of the ML (Engineered vs natural).
- Remarkable features to enhance prediction accuracy were discussed.
- Future dictions were proposed to overcome the limitation of ML in bioprocess.

GRAPHICAL ABSTRACT



ARTICLE INFO

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Engineered system
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ABSTRACT

Recent advances in machine learning (ML) have revolutionized an extensive range of research and industry fields by successfully addressing intricate problems that cannot be resolved with conventional approaches. However, low interpretability and incompatibility make it challenging to apply ML to complicated bioprocesses, which rely on the delicate metabolic interplay among living cells. This overview attempts to delineate ML applications to bioprocess from different perspectives, and their inherent limitations (i.e., uncertainties in prediction) were then discussed with unique attempts to supplement the ML models. A clear classification can be made depending on the purpose of the ML (supervised vs unsupervised) per application, as well as on their system boundaries (engineered vs natural). Although a limited number of hybrid approaches with meaningful outcomes (e.g., improved accuracy) are available, there is still a need to further enhance the interpretability, compatibility, and user-friendliness of ML models.

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1. Introduction

A biological process (bioprocess hereafter) is a process that utilizes the metabolism of living cells of microbes (e.g., bacteria, archaea, protozoa, algae, fungi, and viruses) and their components (e.g., enzymes) to obtain value-added products, which involves a variety of processes, including both anthropogenic activities (e.g., cultivation, fermentation, and production of pharmaceutical products such as antibiotics) and broadly natural phenomena, such as degradation of organic matter, algal bloom, and nitrogen fixation. Because such living organisms are sensitive to surrounding conditions, the performance of bioprocesses depends on numerous factors essential for living, such as temperature, pH, oxidation–reduction potential (ORP), micronutrients, and coexisting substances, which can be either promoters under optimal conditions or inhibitor inducing stress under extreme conditions (Cho et al., 2018; Seo et al., 2022, 2021).

Despite the limited applications of local biostimulation and bioaugmentation (Azam and Finneran, 2013; Banet et al., 2021; Ouriache et al., 2020; Ramadass et al., 2018; Woźniak-Karczewska et al., 2019), bioprocesses in nature are often regarded as uncontrollable because of their large-scale and invariant nature (e.g., reliance against various interferences). Thus, in most cases, adaptation has long been a prior task in an attempt to make better predictions, such as meteorological forecasting (Mishra and Desai, 2006). In contrast, numerous efforts have been made to enhance the productivity of anthropogenic bioprocesses. Fermentation is a representative example of a controlled bioprocess, and the production of cheese or other dairy products is a delicate outcome of an optimized serial process (Koutinas et al., 2009; Magalhães et al., 2010). Anaerobic digestion (AD) of organic wastes also requires precise control for stable operation (Brown and Li, 2013; Cho et al., 2013; Zhou et al., 2018). Similarly, various types of biological treatments are applied in organic waste (and wastewater) treatment, including composting and conventional activated sludge systems with controlled aeration schemes to increase the treatment efficiency (An et al., 2022; Chung et al., 2014; Kim et al., 2022; Miao et al., 2017; Pan et al., 2015; Pleasant et al., 2014; Wang et al., 2011). Although the aforementioned examples have a lot in common, in addition to utilizing microbial activity, different features (e.g., the scale of the system, species of interest, various promoters, and inhibitors within surrounding matrices) under various applications make it difficult to operate bioprocesses and estimate their performance owing to numerous direct and indirect factors, which become more in a coupled (e.g., hydro-biochemical) process (Ajami and Gu, 2010; Kang et al., 2019; Seo et al., 2021).

As specific processes become more sophisticated, the monitoring and control factors required for process operations increase. For example, a variety of electrochemical, chromatographic, and spectroscopic equipment have frequently been used for monitoring and automatic process control of complex bioprocess reactors, such as AD systems (Nguyen et al., 2015). However, the numerical model based on physical or biochemical theory (i.e., process-based model) has a limitation in simulations when there are many factors to be considered simultaneously. For example, several modeling frameworks that can partially simulate membrane bioreactors (MBR) with membrane fouling are available, but integrated modeling applicable to whole MBR processes has not yet been developed because of its complexity (Zuthi et al., 2012). As the number of input variables increases or big data becomes available, the possibility of applying data-driven models based on the correlation between input and output data has increased (Bzdok et al., 2017). One representative data-driven model is machine learning (ML), which has been widely used successfully in a variety of application domains in recent years owing to the rapid increase in computational resources and the development of various supporting algorithms (Jordan and Mitchell, 2015). ML utilizes an algorithm that learns from the relationship between the input and output data to establish a model. Then, the developed ML model produces new predictions (numerical values or labeled data) using the fresh input values (Jordan and Mitchell, 2015).

It has also been actively applied in the field of biology. In particular, active research is being conducted to understand and predict protein structures (AlQuraishi, 2021), which have been an enigma for a long time owing to their complexity. ML algorithms have also been applied to bioprocess modeling, which has limitations in simulating complex microbially mediated physicochemical mechanisms in bioprocesses (Wang et al., 2020a). For example, the anaerobic digestion model (ADM) has been developed to predict process performance, and the activated sludge model (ASM) has been employed to estimate the quality of produced water from wastewater treatment processes (WWTPs) (Batstone et al., 2002; Henze et al., 2000). However, their limitations were continuously pointed out in reported studies that conventional process-based models were unable to accurately predict complicated biological activities (Cruz et al., 2022; Sin and Al, 2021). As an alternative, ML has been applied for the process optimization of complex biological phenomena or prediction of specific bioproducts, such as biofuel/biogas, in both engineered and natural systems (Cruz et al., 2022; Guo et al., 2021; Jeong et al., 2021; Seo et al., 2021; Wang et al., 2020). These bioprocess data often have time-series characteristics; thus, deep learning (DL) techniques specialized for time series forecasting have been effectively applied for developing time-dependency models in biological data (Gopakumar et al., 2018; Krishna et al., 2021; Natarajan et al., 2021). For example, a study employing an integrated mechanistic and DL model was recently proposed to predict the enrichment of polyhydroxyalkanoate-accumulating bacteria in mixed microbial cultures (Xu et al., 2022). Representative DL techniques often include recurrent neural networks (RNN), long short-term memory (LSTM), transformer (TF), and their variants such as a combination of these models using ensemble methods (Seo et al., 2021; Son et al., 2021; Son et al., 2022; Vaswani et al., 2017). In addition, an explainable artificial intelligence (XAI) such as Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) have been introduced to facilitate the interpretation of any developed ML models which are not readily understood in conventional models (Yoon et al., 2021).

Despite its numerous advantages, there are still several challenges in applying ML to the practical application of bioprocesses. The first issue is the quantity and quality of the biological data (Charaniya et al., 2008). It takes a large amount of carefully labeled data to develop a successful model, and the model only works correctly when the data used for model training and the data collected in the actual application are highly similar. This is also related to the issue of sensors because the types of sensors that can measure meaningful data regarding microbial activities are still limited (Alford, 2006). The second issue is the versatility of the proposed model. As aforementioned, the model is expected to function most effectively when the data used for the actual application (i.e., the test step) are similar to the data used for the training step. In other words, it is challenging to immediately adapt a model discovered from the data from one plant to another, even within the same process. Transfer learning performs this type of job; however, there are no strict guidelines to implement it for the bioprocess in a scalable and flexible manner owing to its delicate metabolic interplay. Finally, ML algorithms have inherent limitations owing to their characteristics in learning patterns from the data. ML learns the correlation between data and labels; in other words, it requires effort to create the correct label with a set of specific data.

Within this context, the present study attempted to delineate various categories of ML applications in bioprocesses in Section 3 from different perspectives. And some unique methodologies to strategically enhance the prediction accuracy are presented in Section 4. Finally, research needs and future directions are further discussed in Section 5.

2. Review methodology

2.1. Keywords and co-occurrence network

Research articles were collected by filtering the research using the keywords “machine learning” or “deep learning” in their “bioprocesses” or “biological processes.” The natural systems were searched using the keywords “algae,” “cyanobacteria,” “macroinvertebrates,” “salmonella,” “phytoplankton,” and “bloom-forming,” which could represent the bioprocess in the ecosystem (1,629 papers). Conversely, for the engineered system, keywords related to treatment technology or product of the bioprocess were used: “anaerobic digestion,” “biofuel,” “biochar,” “biomass,” “biofuel,” “pyrolysis,” and “micropollutants” in “water and wastewater treatment process” (508 papers). Notably, between 2000 and 2002, the number of works covering those topics increased exponentially.

The keywords discussed in the title, abstract, and keywords of the collected papers were analyzed using the VOSviewer software (version 1.6.18) (Fig. 1). The size of the circle represented the frequency with which the keywords appeared in the papers. Concerning the natural system, 1,629 papers were collected and deep learning was more frequently employed in their study than machine learning. The research area comprised an entire ecosystem; additionally, the Internet of things and remote sensing were utilized in their studies. On the other hand, machine learning was more frequently used in the 508 papers that focused on the engineered system. Many studies had an interest in biomass, pyrolysis, and anaerobic digestion.

2.2. Systematic review process

Fig. 2 conceptualizes the overall workflow of the study. Numerous literature articles, including research papers and reports identified through Scopus and Google Scholar, were reviewed, and 110 studies (specifically, 7 from the 2000 s, 39 from the 2010 s, and 59 from 2020 to the present) were selected. These dealt with the following: (1) recent trends in ML/DL application to bioprocesses, (2) their categorization from different perspectives, and (3) unique attempts to enhance the prediction accuracy. The specific methodology adopted in each section is as follows.

- Categories of ML applications in bioprocesses were first explored in Section 3 regarding the classification of various ML applications to bioprocess with ML types (i.e., supervised and unsupervised learning), after which they were further classified with system boundaries both in scale- (i.e., engineered vs natural system) and applied location- (surface vs subsurface) perspectives.
- Based on the literature, the origin of inherent uncertainties is selectively reviewed and presented in Section 4, and some unique technologies from recent research articles to strategically overcome these are also introduced.

3. Categories of machine learning applications in bioprocesses from different perspectives

Because numerous types of ML can be applied to a broad range of bioprocesses, proper classification should be performed depending on the purpose of the ML per application (i.e., supervised vs unsupervised), as well as on their system boundaries (engineered vs natural), which will be further classified and discussed in the following sections.

3.1. Proper methods upon the application: Supervised and unsupervised learning

3.1.1. Regression and classification in supervised learning

Among the ML approaches, supervised learning is a method in which outputs are provided according to the input data (Sen et al., 2020). Supervised learning can be further divided into two categories: regression and classification. Regression creates a model (e.g., function) for continuous data that explains the relationship between dependent and independent variables. In contrast, functions can be established in classification to infer categorical variables. Among the supervised ML algorithms, artificial neural networks (ANN), support vector machines (SVM), k-nearest neighbor (KNN), and random forest (RF) algorithms have been frequently employed in bioprocesses, depending on the objectives of previous studies (Li et al., 2022; Wang et al., 2020a; Xing et al., 2019) (further information available from the summary table in Section 4).

Unlike conventional process-based models that have a unique governing equation for a specific problem with constitutional relationships (i.e., analytical models), various ML models can be adopted for a given

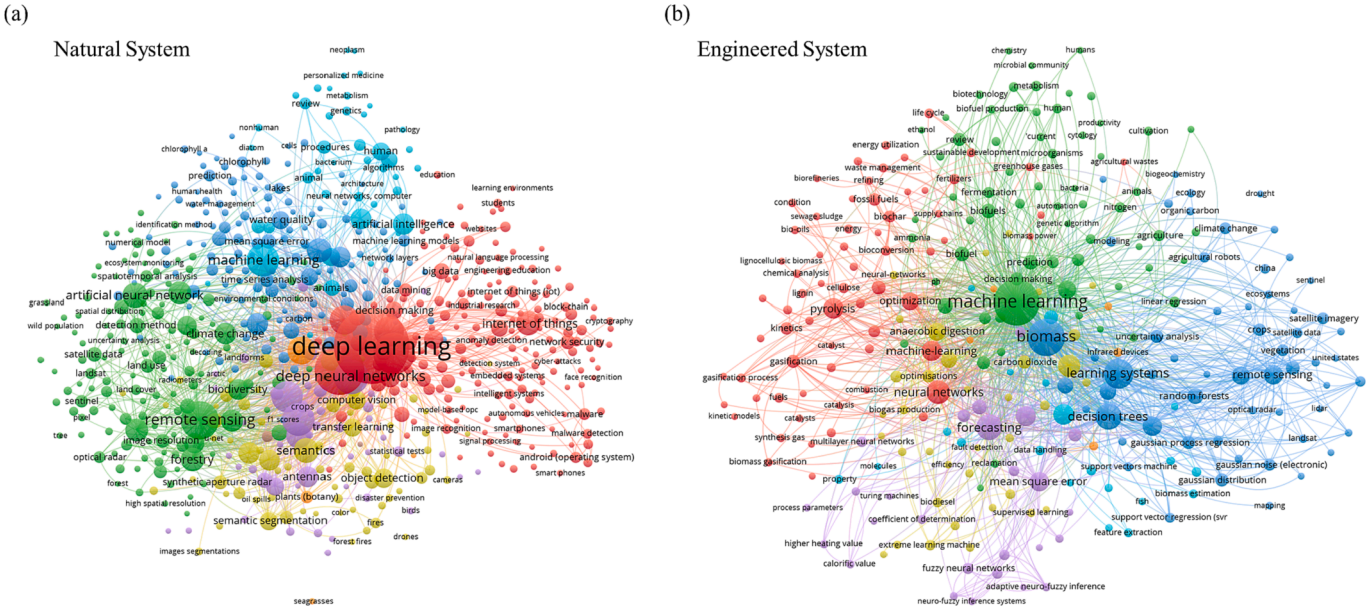


Fig. 1. The keywords occurrence and network of the published papers employing machine learning or deep learning approach in bioprocesses for (a) natural and (b) engineered systems. VOSviewer software was used to visualize bibliometric co-occurrence networks.

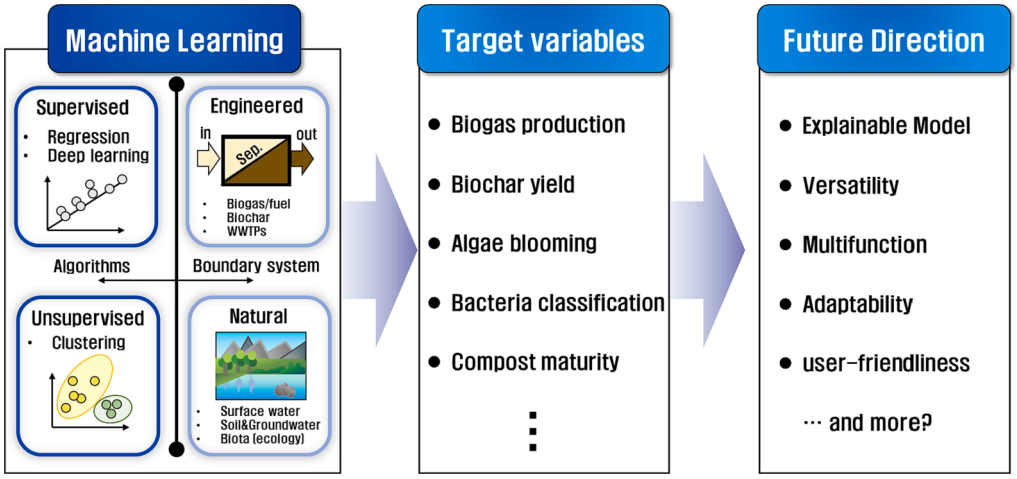


Fig. 2. Schematic description of the overall workflow used in the study.

bioprocess. Therefore, comparisons between ML models using different algorithms have often been conducted to evaluate their prediction performance. Wang et al. (2020a) conducted regression and classification of methane production in the AD process using RF, logistic regression multiclass (GLMNET), SVM, and KNN. Among them, KNN showed the best performance for estimating the amount of biogas produced. In contrast, classification could be effectively performed using GLMNET to predict biogas production in three equal-width groups with a 0.73 of classification accuracy. The critical operation parameters were also determined using RF. Alejo et al. (2018) compared various ML algorithms to predict the effluent composition in a two-stage AD process using SVM and ANN. In their study, SVM showed a higher performance than ANN. In addition to ML models, DL models have recently been applied to predict the performance of dry AD processes. Seo et al. (2021) predicted the biogas production rate using an RNN, which is specialized for time-series data prediction. The biogas production rate of dry AD was predicted using the solid retention time, soluble chemical oxygen demand, total volatile fatty acids, total ammonia, and free ammonia as input features. The coefficients of determination for training and test datasets were 0.82 and 0.85, respectively. Similarly, Jia et al. (2022) employed DL algorithms to predict the state and performance variables of the AD process (e.g., alkalinity, volatile fatty acids, organic matter, and methane production) using pH, electric conductivity, and oxidation and reduction potential data obtained from online sensors as input data. They combined a convolutional neural network (CNN) and LSTM. A CNN was used to extract features from the sensor data, and LSTM was then utilized to predict the performance variable. The combined CNN-LSTM model predicted the performance variable with a 0.978 of R^2 value.

ML algorithms have also been applied to biological WWTPs to predict the influent and effluent characteristics. Zhou et al. (2019) employed the RF algorithm and proposed a probabilistic prediction approach, where weather features and domestic water usage patterns were collected as input features to predict the influent characteristics of two WWTPs in Canada, and the prediction accuracies (R^2) were found to be 0.722 and 0.584, respectively, for the test dataset. Wang et al. (2021) investigated the relationship between operating factors (e.g., influent temperature and dissolved oxygen) and effluent parameters (total suspended solids (TSS) and phosphate (PO_4) in WWTP effluent). They proposed RF and deep neural network (DNN) algorithms to build a model with variable importance measures and partial dependence plot analyses to interpret the model results. The applicability of their ML framework was evaluated (i.e., validated) using operational data from the Umeå WWTP in Sweden. In their study, both RF and DNN could predict TSS and PO_4 with an accuracy higher than 0.87 of R^2 for the test dataset.

In addition to applying ML techniques to continuous numeric data, it can be used for bioimage recognition. Zhu et al. (2021) proposed rapid, accurate, and noninvasive ML approaches to detect biomass and plant cell browning of *Siraitia grosvenorii* cells. The ML approaches consisted of a classification model for cell browning and a regression model for predicting biomass. Different ML models (KNN, SVM, Gaussian process (GP), Decision tree (DT), Naive Bayes (NB), Adaboost, RF, and ANN) were employed. In the feature extraction step, the image dataset was processed into numerical datasets comprising the mean of red, green, blue (RGB), luminosity (the sum of red, green, and blue) with relative color, and hue, saturation, and value (HSV). Eight ML models could classify the brown cells and nonbrown cells with ca. 97 % accuracy. In the prediction step, linear SVM showed the best prediction accuracy with 0.0194 as the root mean square error of prediction. On the other hand, Xue et al. (2019) proposed an evaluation method for compost maturity using the DL model. CNN, especially the ResNet model, was employed to classify the compost maturity. Thirty thousand images were collected and divided into four datasets. Four types of materials, vegetable waste, straw, livestock manure, and their mixture, were used, and their maturity was classified into immature or mature. They compared the prediction accuracy of ResNet with that of other CNN models, such as AlexNet, VGG 16, and VGG 19. The ResNet-based model could quickly and precisely predict the compost maturity compared to others. The proposed ML approaches could predict the compost maturity with accuracies of 99.7 %, 99.4 %, 99.7 %, and 99.5 % for their four test datasets.

In supervised learning, simple data with low-level complexity could be predicted using simple neural networks or tree-based models, while complex data were predicted with more complicated neural networks, such as DL. However, a universal model capable of making all the predictions with one model has not yet been developed owing to the different characteristics of the data used.

3.1.2. *Clustering in supervised learning*

Unlike supervised learning, unsupervised learning is employed for data in which correct answers are not given according to the input data (Ghannam and Techtmann, 2021). Therefore, unsupervised learning includes algorithms for clustering and association rules to explore the data and identify subgroups and similarities in the dataset. The subgroups found in the clustering analysis can be used to set the classification criteria for the input data (Chakraborty, 2017). For example, hierarchical clustering analysis has recently been employed in WWTPs. The consensus for driving the biodegradation of micropollutants is lacking because the physicochemical properties of micropollutants and

process parameters of WWTPs have limitations in fully explaining the biodegradation of structurally different micropollutants (Wang et al., 2020b). Therefore, ML approaches have been used to analyze the degradation pathways of micropollutants to overcome the limitations. Achermann et al. (2018) performed a clustering analysis using the rate constant obtained from batch experiments and analyzed the effects of solid retention time on the biodegradation of micropollutants. Gallé et al. (2019) performed a clustering analysis by elimination rate (i.e., the normalized values of flow and clogging influences for target micropollutants using those of reference compounds, such as carbamazepine or lidocaine). The results of clustering analysis according to the elimination rate were analyzed using biotransformation rules, which predict possible biotransformation pathways depending on the chemical structure of micropollutants (Ellis and Wackett, 2012; Ellis et al., 2008). Wang et al. (2020b) conducted a clustering analysis using biotransformation rules to improve the prediction of micropollutant removal efficiency. Therefore, a regression analysis was conducted after clustering to increase the prediction accuracy of the removal efficiency. However, the prediction accuracy ranged from 0.09 to 0.75, depending on the cluster. Although biotransformation rules were established based on chemical structures, they could not consider every possible microbial reaction pathway determined by the presence of functional groups within the micropollutants. As aforementioned, the ML-based clustering techniques are shown to be effective in classifying complex data in bioprocesses, which could be combined and further processed with another model (e.g., regression) for better performance.

3.2. System boundary: engineered vs natural system

3.2.1. Bioprocesses in an engineered system

In the engineering field, ML algorithms have been employed in process control owing to their estimation and prediction capabilities. Most of the aforementioned studies in **Section 3.1** focused on the prediction of process performance (Alejo et al., 2018; Seo et al., 2021; Wang et al., 2020a). Therefore, their targets were the amount or rate of production, that is, the process performance. Process control can only be achieved by accurately estimating or predicting the state and performance variables of a real-time process. Jia et al. (2022) used a combined CNN-LSTM model to predict TSS and PO_4, and the model could predict them with an 0.98 of R^2 value. It was believed that ML approaches could support decision-making for optimal control strategies. Yang et al. (2021) proposed a real-time control strategy for combined alkaline and ultrasonication sludge lysis-cryptic pre-treatment and anoxic/oxic systems using an ANN. Their control strategy was established based on the accurate regression performance of the ANN model with a 0.95 R^2 value.

Critical operational parameters should be determined to effectively control the bioprocess. For example, ML has been applied to control the amount of biochar yield and specific surface area with total pore volume to produce a good quality biochar, having great versatility in various environmental applications, such as soil amendment and as an adsorbent for removing harmful substances both in air and water. Therefore, previous studies have employed RF to predict process performance and evaluate the feature importance of input features (Leng et al., 2021; Pathy et al., 2020; Zhu et al., 2019). Alternatively, sensitivity analysis is also applied to determine the critical operating parameters among the input features of the ML models as a post-hoc analysis. Zhao et al. (2021) utilized the kernel extreme learning machine (KELM) and the Kriging model to predict heavy metal removal using biochar. They then conducted a local sensitivity analysis to evaluate the relationship between the input features and the adsorption capacity of biochar. Among the 15 influencing factors, pH and temperature showed a strong correlation with adsorption efficiency in their study. The ML models used for process control can also be used to estimate the abnormality of the process by comparing sensor monitoring data with the simulation results of the model. Valim et al. (2017) proposed an ANN model for fault detection during the delignification of sugarcane bagasse. The ANN model was used to predict the percentage of lignin oxidized from bagasse. The prediction accuracy of the ANN model was greater than 90 %. They used ANN models to identify faults by comparing the predicted data from the ANN with observed data. Seo et al. (2021) applied a combined RNN and LSTM to predict the system failure of continuous dry AD operations.

From the above examples, applying ML in an engineered system can have the following advantages: 1) Prediction of process performance with high accuracy, 2) fault detection, and 3) decision-making support for optimal control strategies. Therefore, further study of these advantages in the future may enable control or automation of the engineered biosystem processes using ML algorithms.

3.2.2. Bioprocesses in the natural system

ML algorithms have been successfully employed for the predictive modeling of complex natural environments, such as surface and subsurface systems. ML applications in natural systems can be further divided into four main categories based on the environmental media: surface waters and biota (for surface systems), as well as soils and groundwater (for subsurface systems).

3.2.3. Surface water

In terms of surface waters, ML was widely adopted for predicting water pollution outbreaks caused by biological pollutants, such as fecal coliforms and algal blooms. Baek et al. (2022) used hierarchical DL, which combined the classification transformer model (for determining the onset date of occurrence) and regression transformer models (for estimating the cell concentration) for plankton, and found that the predicted results agreed well with the observed ones regardless of the type of plankton. Derot et al. (2020) showed that the coupling of K-means and RF models could forecast toxic bloom-forming *P. rubescens* in a particular lake. El Bilali et al. (2021) combined adaptive boosting (Adaboost) models with a Gaussian noise-based data augmentation method to avoid the overfitting problem in predicting fecal coliforms at riverine sites with a small dataset. An integrated approach based on ML algorithms and decomposition methods was also examined to predict cyanobacterial blue-green algae in the study by Heddam et al. (2022), which demonstrated that the decomposition methods helped increase the predictive accuracy of individual ML. Hussein et al. (2019) reported that a hybrid method connecting the moth search algorithm to the random vector functional link network was suitable not only for recovering the missing values of the total algal count but also for improving the performance of the original model. Note that the proposed hybrid method plays a role in minimizing the number of input variables from 34 to 8, and eventually to 4, and reducing the processing time. Mohammed et al. (2022) documented that the LSTM algorithm accurately predicted *E. coli* and fecal coliform levels at different depths of a specific lake generated from the process-based model simulation. The predictive accuracy of microcystic occurrence was tested in a dataset where the number of variables was larger than the number of observations, and the feature engineering algorithm contributed to accelerating the performance of ML (Mori et al., 2022). Pahlevan et al. (2020) adopted a mixture density network that outperformed other state-of-the-art algorithms in retrievals of chlorophyll-a concentrations, and found that the model performance was significantly enhanced over existing algorithms, irrespective of satellite images obtained in inland and coastal waters. Pyo et al. (2020, and 2021) compared the performance of a CNN with and without a convolutional block attention module and revealed that both models effectively estimated cyanobacteria concentrations in river waters.

3.2.4. Biota in surface systems

ML was also applied to address the geographical distribution of biological organisms, including fish and aquatic macrophytes, and the ecological determinants of species occurrence. Cha et al. (2021) predicted the occurrence of freshwater macroinvertebrates using RF and then used SHapley Additive exPlanations (SHAP) to make the developed

model more explainable. The incorporation of SHAP into the ML model, in particular, identified nonlinear species responses to environmental variables, as well as the existence of ecological thresholds. A study by da Silveira et al. (2021) showed that the SVM classifier was suitable for habitat mapping of coral reefs and that human activities were responsible for the inhibition of coral development. Dashpurev et al. (2021) reported that vegetation changes were successfully captured by the use of machine learning in aerial imagery of any resolution in a cost-effective manner, but with higher accuracy for unmanned aerial vehicle imagery with 2 cm resolution than for PlanetScope imagery with 3 m resolution. The invasion risk and its determinants for freshwater fish were studied using an ensemble species distribution model, which confirmed that the developed model was effective in delineating the areas of concern and identifying factors affecting future invasion (Dong et al., 2020). Espel et al. (2020) assessed macrophyte distribution in rivers by applying ML algorithms to several spectral datasets and found that RF combined with the four spectral bands performed better than the other one, even in extreme (macrophyte) cover values (i.e., 0 % and 100 %). Henriques et al. (2022) combined multispectral and radar imagery data to create habitat maps of migratory shorebirds using ML algorithms, and demonstrated success in improving their performance in the presence and absence of this combined dataset. Li et al. (2019) adopted three supervised classification algorithms to detect mangrove changes, and Sharma et al. (2021) used ensemble models to elucidate the current suitability and future range shifts of snow trout. Pichler and Hartig (2021) introduced a new joint species distribution model that is much faster and more accurate than the original model in describing species association structures. In addition, Tabak et al. (2019) trained a CNN model with the ResNet-18 architecture for classifying wildlife species from camera trap images and achieved high prediction accuracy. Several studies have assessed the aboveground biomass of mangrove plants (Tian et al., 2022) and developed vegetation maps using a series of ML algorithms (Wu et al., 2021).

3.2.5. Subsurface systems: Soil and groundwater environments

ML has received little attention for subsurface systems. With respect to soils, Mori et al. (2019) studied the interactions between environmental factors and biologically driven processes (i.e., respiration and microbial respiratory electron transport system activity (ESTA)) that occur in the hyporheic zone using decision trees and found that the predictive accuracy for respiration was considerably higher than that of ESTA. Note also that these interactions are heavily affected by the environmental variables selected for (target) variables of interest. Spahn et al. (2022) adopted DL for multitask bacterial image analysis, more specifically *B. subtilis* for soil samples and *E. coli* for groundwater samples, and revealed that different DL approaches quickly allowed for image segmentation of different bacterial species, classification of different growth stages, and accurate mapping of cell shape and intracellular targets in series. The predictive performance of the target standards classified into two levels based on the *E. coli* concentration was also examined using ML algorithms (Tousi et al., 2021). Note that SVM performed the best, and *E. coli* concentration in the sediment was selected as one of the contributing factors responsible for the changes in the condition of irrigation water.

Finally, ML was used for the detection of microbiological contamination (e.g., fecal coliforms and *Salmonella*) in groundwater. Díaz-Alcaide and Martínez-Santos (2019) reported the usefulness of various ML classifiers in developing fecal pollution maps in rural groundwater supplies and suggested them as credible alternatives to traditional Darcian approaches. Bedell et al. (2022) applied ML to fluorescence sensor data to continuously assess fecal contamination risk in surface waters; however, the developed model was eligible for contamination prediction for groundwater even in the presence of biofilm formation on the sensor lenses. Rocha et al. (2022) adopted non-parametric supervised ML (with *meta*-regression) to examine the relationship between moderators and the occurrence of *Salmonella* in surface waters and groundwater and revealed that the source of water was strongly related to the frequency of *Salmonella*. White et al. (2021) aimed to identify the drivers of *E. coli* contamination in private wells using ML as well as the relative importance of independent variables and demonstrated that the proposed methodology was helpful in describing the potential risks associated with the use of private wells.

Overall, ML models are widely used to predict bioprocesses in the natural system including algae in surface water, the geographic distribution of specific species in biota, and predict *E. Coli* and other microorganisms in the subsurface. Among them, the most up-to-date ML model (i.e., hybrid or transformer) is used in the field of biota presumably due to their complex mutual interaction, where there is a need for processing high-dimensional (or large-scale) data.

4. Research needs and future direction

ML approaches in bioprocesses can be effectively implemented by analyzing the originality and shortcomings of previously reported studies. The selected studies are summarized in Table 1, with an additional focus on remarkable approaches to reduce the inherent uncertainties from the applied ML models. The studies were categorized depending on the types of algorithms and system boundaries within the applied research fields, and then investigated alternative methods employed to overcome the limitations presented previously. In addition, a method to deal with the interpretability of ML algorithms, one of the most controversial issues in ML approaches, was investigated. The research requirements are discussed in Section 4.1, and future directions for the application of ML algorithms in bioprocesses are proposed in Section 4.2.

4.1. Research needs

A limitation that prevents the wide application of ML approaches is their interpretability (Rudin, 2019). This is why a tree-based machine learning algorithm (e.g., RF and extreme gradient boosting) have been utilized in recent studies (Ascher et al., 2022; Wang et al., 2020a). To further supplement this, ML approaches might be directly applied to estimate the activities of microbial communities, in addition to routinely measured operating parameters. For example, it was previously pointed out that a proper analysis of microbial communities should be performed to overcome the limitations of conventional ASM, which merely considers the complicated bioreactions in the activated sludge process as simple degradation (Sin et al., 2021). Sin and Al (2021) also proposed that the overall estimation of bioprocesses can be enhanced by applying the ML algorithm to process biological data regarding microbial communities, which has rarely been considered in conventional models. In addition to WWTPs, microbial communities play an important role in the anaerobic processing of organic waste (Blasco et al., 2022; Ketsub et al., 2022). Although previously conducted studies showed high prediction accuracies in bioprocess performance, their models were lumped and black-box models, which partially consider biological communities. If biological information could be fully utilized as input data, it could reduce the uncertainty originating from complicated bioreactions. In addition, versatility must be satisfied for the widespread use of the developed model. Transfer learning, which involves performing fresh learning based on an already taught model, or models that include updating the learning of a model based on real-time input, is being developed as a supplement to this (Zhuang et al., 2021).

4.2. Future direction

Owing to the rapid advancement of ML algorithms, several MLs have been successfully adopted in bioprocesses, as described above. However, several issues still need to be overcome to facilitate the use of ML in bioprocesses for accurate prediction of the desired outputs (i.e., biogas production) or even for automation of the systems. Finally, learning is a

Table 1
Summary of the important features from the most recent works that applied ML to bioprocess from different perspectives.

Analytical task	System boundary		Target variables	Algorithms [a]	Remarkable features	References
Regression	Engineered	Anaerobic digestor	Methane production rate	RNN	- Time series prediction - System failure (fault detection) - Comparison with a process-based model	Seo et al. (2021)
			Alkalinity, volatile fatty acids, organic matter, and methane production	Combined CNN and LSTM	- CNN for feature extraction and LSTM for prediction - Evaluation of generalizability for the proposed model (further study required)	Jia et al. (2022)
		Adsorption batch reactor (biochar)	Removal efficiency of heavy metal	KELM and Kriging model	- Employment of local sensitivity analysis to evaluate the relationship between input features and targets	Zhao et al. (2021)
	Natural	Surfacewater	Cyanobacteria concentrations	CNN	- Effect of attention module in the estimation of targets for enhancing model accuracy	Pyo et al. (2020 and 2021)
		Biota in surface systems	Occurrence of freshwater macroinvertebrates	RF	- Explainable model by using Shapley Additive exPlanations	Cha et al. (2021)
Regression and Classification	Engineered	Anaerobic digestor	Methane production	RF, GLMNET, SVM, and KNN	- Determination of critical operation parameters using RF	Wang et al. (2020a)
		Composting	Compost maturity	RF, XGBoost, LightGBM, and MLP	- Explainable model by using Shapley Additive exPlanations - Comparison between single- and fusion-model	Wan et al. (2022)
	Natural	Surfacewater	- Date of occurrence - Types of plankton	Transformer model	- Transformer models for classification and regression	Baek et al. (2022)
Clustering and Regression	Engineered	Wastewater treatment	- Grouping of micropollutant (Clustering analysis) - Removal efficiency (Regression)	Hierarchical clustering and GAM	- First attempt to cluster micropollutants based on biotransformation rules - Limitation of biotransformation rules for reflecting every possible bioreaction	Wang et al. (2020b)
	Natural	Surfacewater	- Bloom-forming *P. rubescens*	K-means and RF	- Combined clustering and subsequent regression approach to forecast harmful algal blooms	Derot et al. (2020)

[a] RF: Random Forest; SVM: Support Vector Machine; KELM: Kernel Extreme Learning Machine; XGBoost: Extreme Gradient Boosting; LightGBM: Light Gradient Boosting Machine; MLP: Multilayer Perceptron; GAM: Generalized Additive Models; CNN: Convolutional Neural Network; RNN: Recurrent Neural Network; LSTM: Long Short-Term Memory.

limitation of ML. ML learns the correlation between data and labels; in other words, it requires effort to create the correct label with a set of specific data. Therefore, reinforcement learning, a learning method through trial and error, has recently been proposed using simple data to address this issue (Petsagkourakis et al., 2020). Based on these, future directions were proposed to apply ML approaches in bioprocess.

First, ML approaches should be more explainable to support decisions made during operations (Gunning et al., 2019). In other words, a limitation that prevents the wide application of ML approaches is their interpretability of ML approaches (Rudin, 2019). This is known as explainable artificial intelligence (XAI) and can overcome the inherent limitation of the black-box model of ML, which has been considered one of the greatest challenges for practical applications (Adadi and Berrada, 2018). In addition, this is the reason that tree-based ML approaches, such as RF and extreme gradient boosting are utilized in recent studies because those are more explainable than other neural networks (Ascher et al., 2022; Wang et al., 2020a). Three representative approaches to XAI in ML are (1) the use of the attention mechanism, (2) feature importance (or permutation importance), and (3) SHAP value (Novakovsky et al., 2022). Although there are differences between these three XAI approaches in a broad framework, they are the same in that they quantify the effect of input variables on output variables. In other words, these techniques can be beneficial for decision-making using ML because it is possible to determine the degree of change in the output variables while varying the input variables.

Second, building a giant model that is trained using a massive dataset is required to develop a multifunctional ML model for practical applications. As the ML model is data-driven, a malfunction can occur if the prediction deviates significantly from the dataset used for training. Conversely, if a large ML model is built, it can make accurate predictions or instructions in most situations. Such an example can be found in a giant ML developed in natural language processing (i.e., GPT-3 (Floridi and Chiriatti, 2020)) or multimodal learning that can perform various tasks by one algorithm (i.e., GATO (Sparkes, 2022)). If such a large ML model is developed for bioprocesses, the uncertainty problem, which is one of the most problematic in prediction using ML, can be partially solved.

Finally, as aforementioned, the ML model should be large and complex but easy to use concurrently. There are two controversial methods for increasing the accuracy of prediction when using ML models. One is to use a proper dataset (feature engineering), and the other is to build a suitable model that can fit well into the dataset (Zheng and Casari, 2018). For example, as the structure of ML becomes more complex, it becomes hard to understand and apply, and thus techniques, such as neural architecture search (NAS (Elsken et al., 2019)) or automation of ML structure search (AutoML (He et al., 2021)) have gained much attention. In other words, it is necessary to develop a highly user-friendly ML model that can work with a small amount of coding, while quickly finding the optimal ML structure for a specific bioprocess.

In this regard, ML can even contribute to the "circular bioeconomy," a concept that includes economic activities connected to biological processes by which waste and byproducts are converted into bioenergy and value-added bioproducts (Akbarian et al., 2022; Kumar Awasthi et al., 2022) by optimizing or ultimately automating the bioprocesses at their highest performance (Tsui et al., 2022).

5. Conclusion

Different types of ML algorithms of supervised and unsupervised learning have recently been actively applied to bioprocesses in engineered and natural systems. Despite their distinct features, most of them are aimed at predicting specific factors with remarkable performance and there seems to be a demand for overcoming the limitations of the conventional models. However, the versatility, interpretability, and ease

of use of ML models are still challenging to solve. The advancement of these models will eventually pave a new way for the wide application of bioprocesses.

CRediT authorship contribution statement

Seung Ji Lim: Methodology, Investigation, Writing – original draft. **Moon Son:** Methodology, Investigation, Writing – original draft. **Seo Jin Ki:** Investigation, Writing – original draft. **Sang-Ik Suh:** Methodology, Investigation. **Jaeshik Chung:** Conceptualization, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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